Dr. D. Y. Patil Educational Enterprises Charitable Trust's

"Empowerment through quality technical education"

AJEENKYA DY PATIL SCHOOL OF ENGINEERING

D Y Patil Knowledge City, Charholi Bk., Via. Lohegaon, Pune – 41205

DEPARTEMNT OF COMPUTER ENGINEERING

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Sr. No.	Title	Page No.	Date	Sign of Faculty	Remarks
1	Understanding Basics of Python Programming and Basic Operations on array				
2	Write a Python program to manage the borrowing records of books in a library. Implement the following functionalities: • Compute the average number of books borrowed by all library members. • Find the book with the highest and lowest number of borrowings in the library. • Count the number of members who have not borrowed any books (denoted by a borrow count of 0). • Display the most frequently borrowed book (i.e., the mode of borrow counts). After performing, determine the time and Space complexity of each operation				
3	In an e-commerce system, customer account IDs are stored in a list, and you are tasked with writing a program that implements the following: • Linear Search: Check if a particular customer account ID exists in the list. • Binary Search: Implement Binary search to find if a customer account ID exists, improving the search efficiency over the basic linear				

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4	In a company, employee salaries are stored in a list as floating-point numbers. Write a Python program that sorts the employee salaries in ascending order using the following two algorithms: • Selection Sort: Sort the salaries using the selection sort algorithm. • Bubble Sort: Sort the salaries using the bubble sort algorithm. After sorting the salaries, the program should display top five highest salaries in the company				
5.	Implementing a real-time undo/redo system for a text editing application using a Stack data structure. The system should support the following operations: • Make a Change: A new change to the document is made. • Undo Action: Revert the most recent change and store it for potential redo. • Redo Action: Reapply the most recently undone action. • Display Document State: Show the current state of the document after undoing or redoing an action				
6.	Implement a real-time event processing system using a Queue data structure. The system should support the following features: • Add an Event: When a new event occurs, it should be added to the event queue. • Process the Next Event: The system should process and remove the event that has been in the queue the longest. • Display Pending Events: Show all the events currently waiting to be processed. • Cancel an Event: An event can be canceled if it has not been processed.				

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7.	Create a Student Record Management System using linked list • Use a singly/doubly linked list to store student data (Roll No, Name, Marks). • Perform operations: Add, Delete, Update, Search, and Sort. Display records in ascending/descending order based on marks or roll number.				
8.	Implement a hash table of size 10 and use the division method as a hash function. In case of a collision, use chaining. Implement the following operations: • Insert(key): Insert key-value pairs into the hash table. • Search(key): Search for the value associated with a given key. • Delete(key): Delete a key-value pair from the hash table				
9.	Design and implement a hash table of fixed size. Use the division method for the hash function and resolve collisions using linear probing. Allow the user to perform the following operations: • Insert a key • Search for a key • Delete a key • Display the table				